

Introduction to Quantum Computing (L-10)

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Grovers Algorithm

- Grover's algorithm is a quantum algorithm designed to search for a specific item in a large list or database
- It is known to be an optimal quantum search algorithm, meaning that no quantum algorithm can solve the unstructured search problem using significantly fewer steps than Grover's method.
- In a classical computer, if you have N possible items, you may need to check about $N/2$ items on average to find the correct one. Grover's algorithm can find the correct item much faster, in about $\sqrt{N}$ steps. This provides a quadratic speedup compared to classical search.
- The algorithm begins by placing the quantum system into a superposition, meaning it represents all possible answers at the same time.
- Applications: Grover's algorithm can be used to speed up database searches, optimization problems, pattern matching, and certain cryptographic attacks (such as reducing the effective security of brute-force key searches).

Grovers Algorithm

Let there be a function $f: \{0,1\}^n \rightarrow \{0,1\}$ which takes 'n' bits as inputs and returns a single bit as output, which is either '0' or '1'. Our goal is to find the specific inputs values for which the function returns '1', that is find $x \in \{0,1\}^n$ such that $f(x) = 1$. Like previous examples, the function is treated like a black box (we are not allowed to look inside)

Examples: $n = 3$

Function domain	Function range
000	0
001	0
010	0
011	0
100	0
101	1
110	0
111	0

Classical algorithm needs to check every input one after another, until it finds a '1'.

$f(000) \rightarrow 0$
 $f(001) \rightarrow 0$
 $f(010) \rightarrow 0$
 $\vdots$
 $f(101) \rightarrow 1$

Classical algorithm complexity is $\mathcal{O}(2^n) \rightarrow$ exponential

Without giving a proof we are stating that in contrast a quantum algorithm complexity is $\mathcal{O}(\sqrt{2^n}) \rightarrow$ polynomial

Grovers Algorithm

Examples:

Function domain	Function range
000	1
001	0
010	0
011	1
100	0
101	0
110	1
111	0

Let $|A\rangle$ be a vector that represents a superposition all inputs that returns '1'

$$|A\rangle = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

If we measure $|A\rangle$, we are supposed to get 000, 011, or 110 with equal probability of 1/3

Let $|B\rangle$ be a vector that represents a superposition all inputs that returns '0'

$$|B\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

If we measure $|B\rangle$, we are supposed to get 001,010, 100, 101 or 111 with equal probability of 1/5

Check that $|A\rangle$ must be orthogonal to $|B\rangle$: $\langle A|B\rangle = 0$

Also check that $\|A\| = \|B\| = 1$

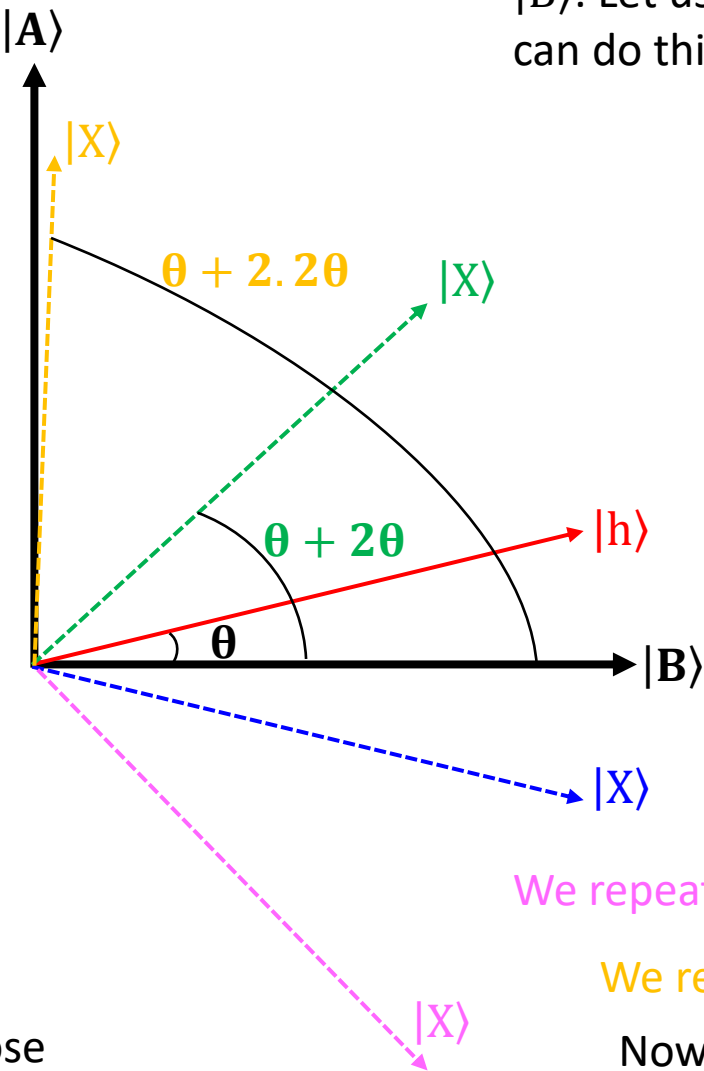
Grovers Algorithm

Example:

Function domain	Function range
000	1
001	0
010	0
011	1
100	0
101	0
110	1
111	0

What is the expected outcome if we measure register $|X\rangle$ at this stage?

At this stage, we are more likely to measure the contents of $|B\rangle$ than those of $|A\rangle$



Let $|h\rangle$ be a vector that represents a superposition of $|A\rangle$ and $|B\rangle$. Let us create a register $|X\rangle$ and initialize it to $|X\rangle = |h\rangle$. We can do this using Hadamard gates

We imagine there is a mirror on $|B\rangle$, and we reflect the register on that mirror

Next, we imagine there is a mirror on $|h\rangle$, and we reflect the new $|X\rangle$ on that mirror

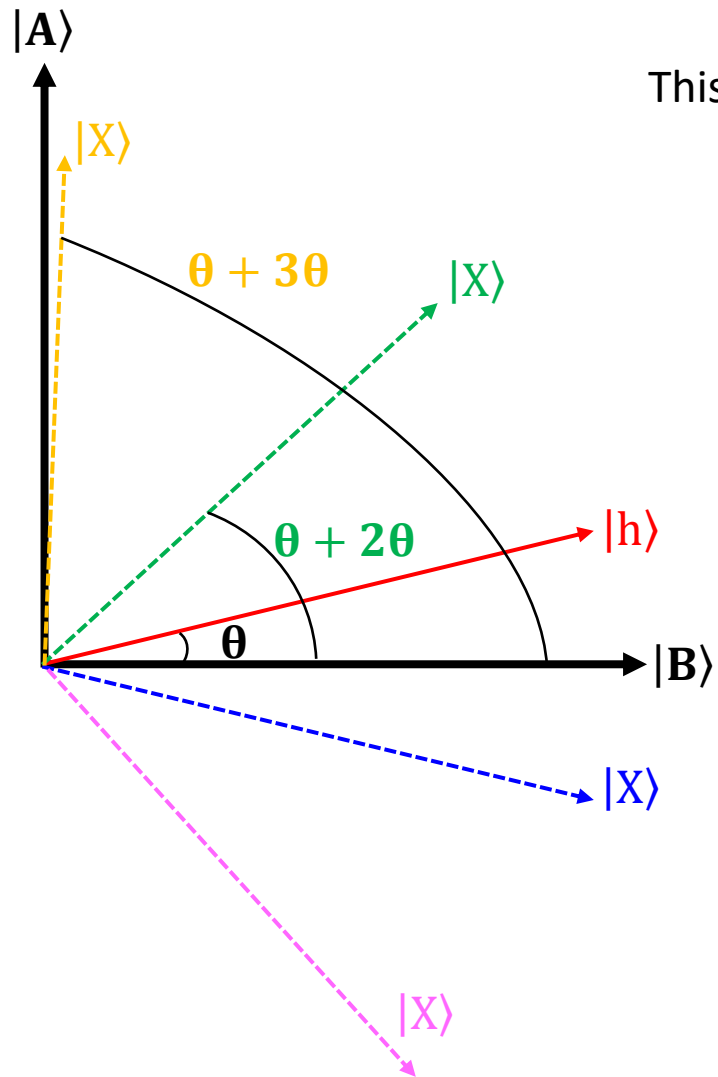
At this stage if we measure $|X\rangle$ we have a greater chance of getting the contents of $|A\rangle$ than before. To enhance its probability, we repeat the process

We repeat the reflection of the new $|X\rangle$ with respect to $|B\rangle$

We repeat the reflection of the new $|X\rangle$ with respect to $|h\rangle$

Now if we measure $|X\rangle$ there is a very high chance of getting the contents of $|A\rangle$

Grovers Algorithm



This is called Grover rotation

Grover algorithm formalism:

1. Create an equal superposition of all possible inputs in $|h\rangle$
 $|X\rangle = |h\rangle$
2. for ($i = 1$ to k)
{
 $|X\rangle = \text{reflect } X \text{ over } |B\rangle$
 $|X\rangle = \text{reflect } X \text{ over } |h\rangle$
}
3. Measure $|X\rangle$

Grover's Algorithm

How many Grover rotations are needed?

x	f(x)
000	0
001	0
010	1
011	0
100	1
101	0
110	0
111	0

$\{A\} = \{010, 100\}$

$|A| = a = 2$

$$|A\rangle = \frac{1}{\sqrt{a}} (|010\rangle + |100\rangle)$$
$$= \frac{1}{\sqrt{2}} (|010\rangle + |100\rangle)$$

$|A\rangle$ is an equal superposition of all the entries in $\{A\}$

$\{B\} = \{000, 001, 011, 101, 110, 111\}$

$|B| = b = 6$

$$|B\rangle = \frac{1}{\sqrt{b}} (|000\rangle + |001\rangle + |011\rangle + |101\rangle + |110\rangle + |111\rangle)$$
$$= \frac{1}{\sqrt{6}} (|000\rangle + |001\rangle + |011\rangle + |101\rangle + |110\rangle + |111\rangle)$$

$|B\rangle$ is an equal superposition of all the entries in $\{B\}$

$|h\rangle$ is an equal superposition of all possible inputs

$$= \frac{1}{\sqrt{8}} (|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)$$

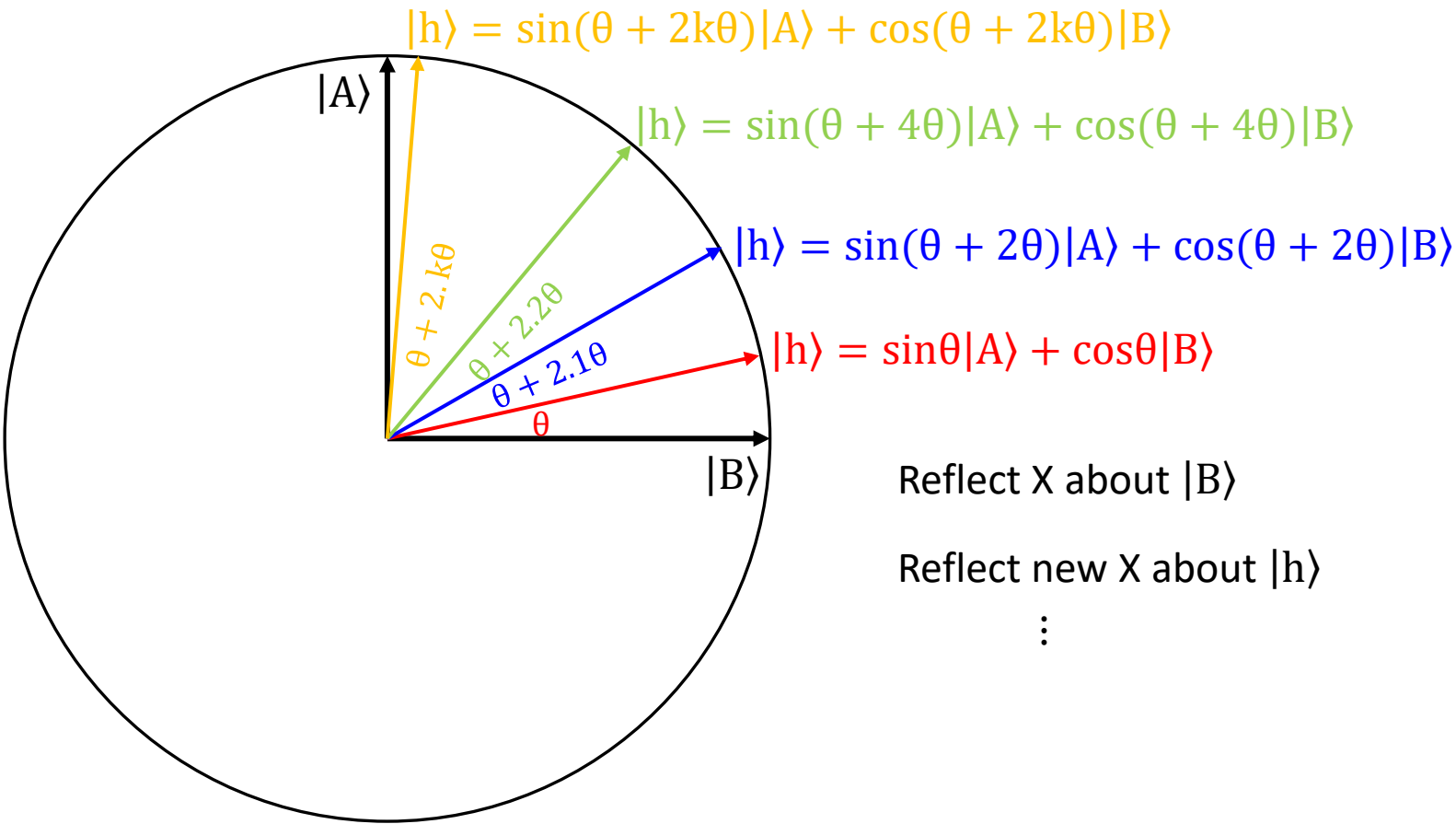
$$|h\rangle = \sqrt{\frac{a}{N}} |A\rangle + \sqrt{\frac{b}{N}} |B\rangle = \sqrt{\frac{2}{8}} |A\rangle + \sqrt{\frac{6}{8}} |B\rangle$$

$n = 3$

$N = 2^n = 8$

Grover's Algorithm

x	f(x)
000	0
001	0
010	1
011	0
100	1
101	0
110	0
111	0



Since we want $|h\rangle$ to be also as close as $|A\rangle$ as possible after 'k' Grover rotation, $|\sin(\theta + 2k\theta)|^2 \approx 1$ and $|\cos(\theta + 2k\theta)|^2 \approx 0$

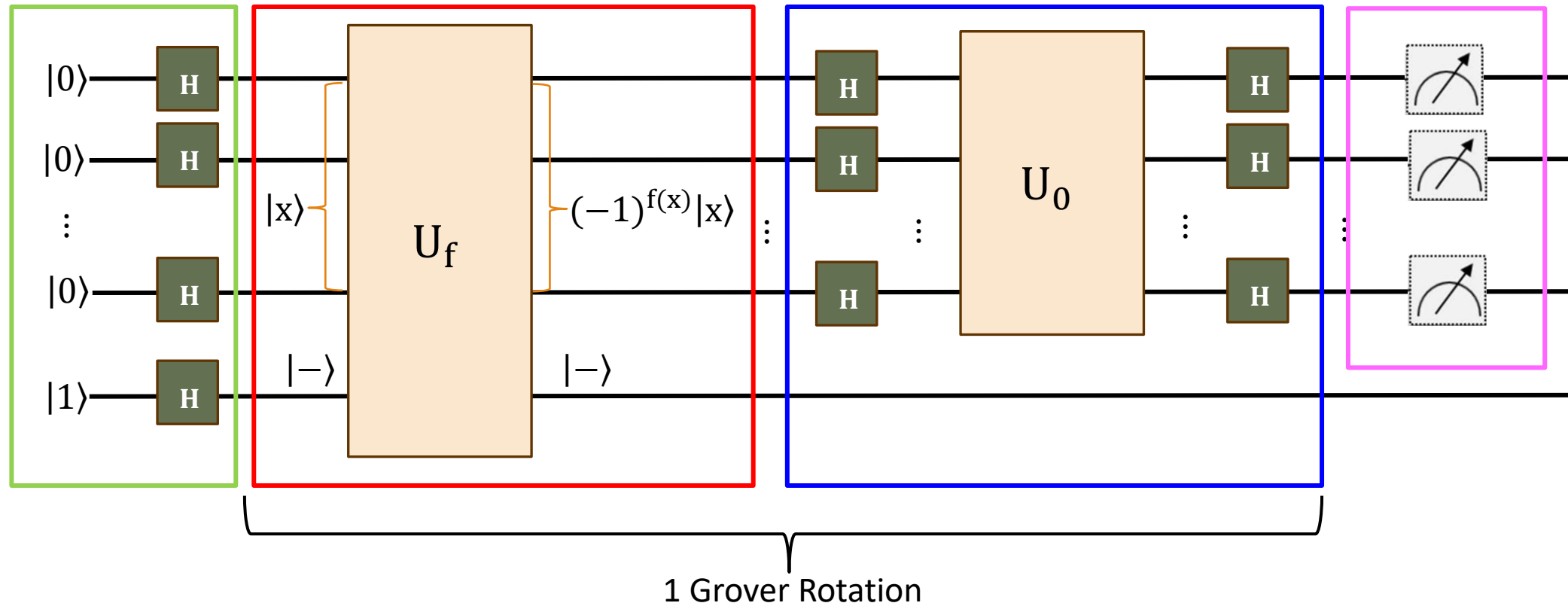
$|\sin(\theta + 2k\theta)|^2 \approx 1 \Rightarrow$

$k \sim \frac{\pi}{4} \sqrt{\frac{N}{a}} - \frac{1}{2}$

$\frac{\pi}{4} \sqrt{8} - \frac{1}{2} \approx 2.22$

$k \sim 2 \text{ (closest integer)}$

Grover's Circuit



1. Create an equal supposition of all inputs and store in X

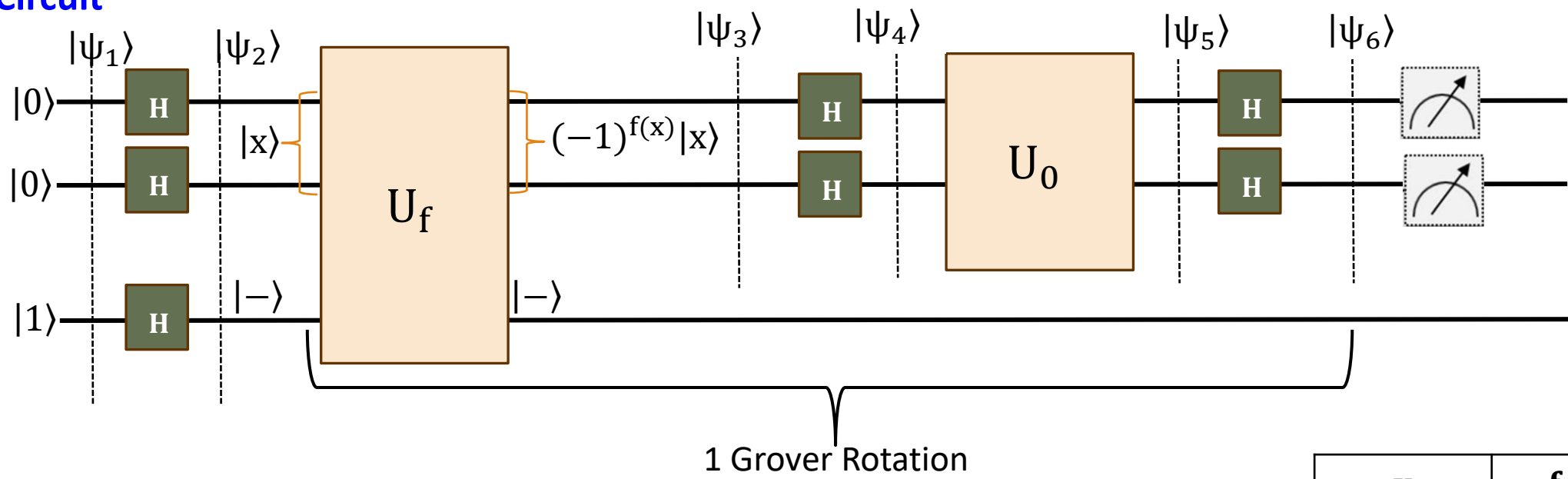
2. Reflect X about $|B\rangle$

3. Reflect X about $|h\rangle$

4. Repeat Grover Rotation 'k' times

5. Measure the first register

Grover's Circuit



How many Grover rotations are needed?

$$k \sim \frac{\pi}{4} \sqrt{4} - \frac{1}{2} = 1.0707 \approx 1$$

$$|\psi_1\rangle = |00\rangle|1\rangle$$

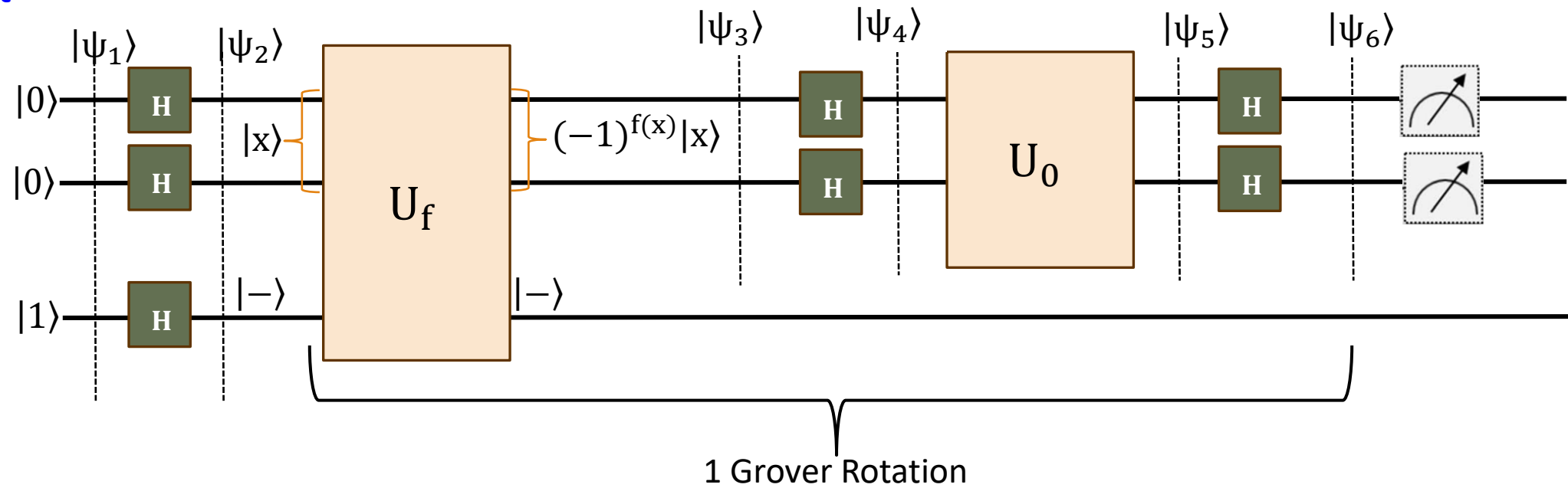
$$|\psi_2\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{\sqrt{2^2}} |- \rangle$$

$$|\psi_3\rangle = \frac{|00\rangle + |01\rangle - |10\rangle + |11\rangle}{\sqrt{2^2}}$$

x	f(x)
00	0
01	0
10	1
11	0

See, we no longer need the second register that is in a $|- \rangle$ state. Its job is done. We can just ignore writing it which is what we are doing here and onward.

Grover's Circuit



$$|\psi_4\rangle = H^{\otimes 2} \frac{|00\rangle + |01\rangle - |10\rangle + |11\rangle}{\sqrt{2^2}}$$

	00	01	10	11
00	+	+	+	+
01	+	-	+	-
-10	-	-	+	+
11	+	-	-	+

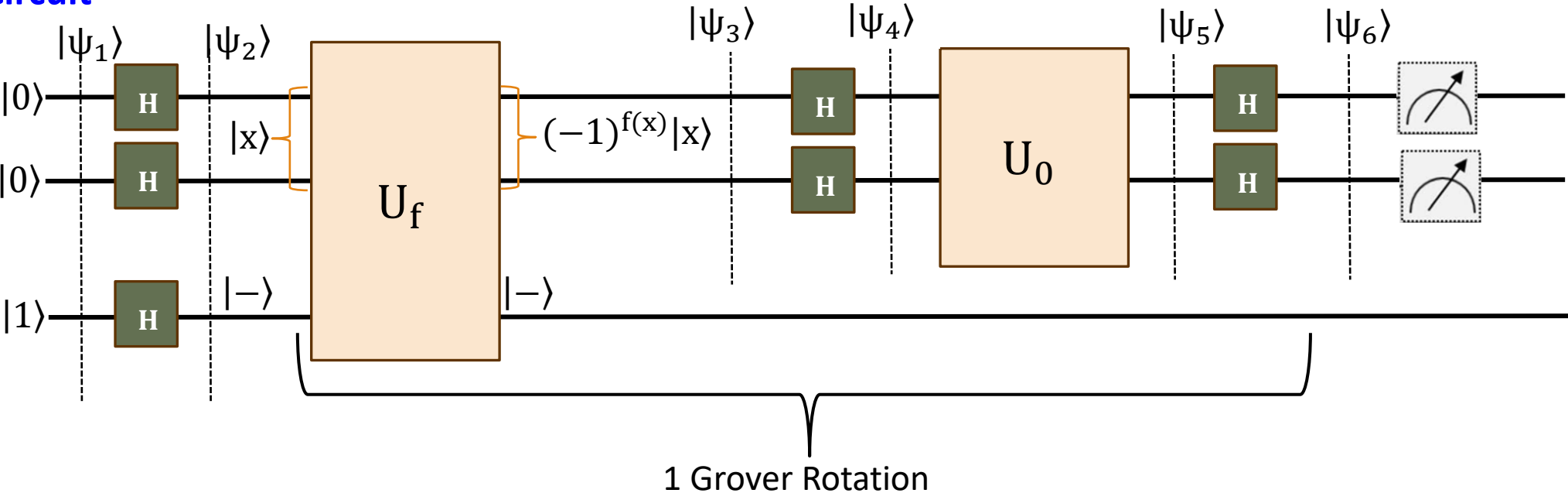
$$|\psi_4\rangle = \frac{2|00\rangle - 2|01\rangle + 2|10\rangle + 2|11\rangle}{\sqrt{2^2}}$$

$$= (|00\rangle - |01\rangle + |10\rangle + |11\rangle)$$

$$|\psi_5\rangle = U_0(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$$

x	f(x)
00	0
01	0
10	1
11	0

Grover's Circuit



$|\psi_5\rangle = U_0(|00\rangle - |01\rangle + |10\rangle + |11\rangle)$

$U_0|x\rangle = |x\rangle$ when $|x\rangle = |0\rangle^{\otimes n}$
 $= -|x\rangle$ when $|x\rangle \neq |0\rangle^{\otimes n}$

$|\psi_5\rangle = |00\rangle + |01\rangle - |10\rangle - |11\rangle$

$|\psi_6\rangle = H^{\otimes 2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle)$

$|\psi_6\rangle = \frac{0|00\rangle + 0|01\rangle + 2|10\rangle + 0|11\rangle}{\sqrt{2^2}}$

Measurement symbol (meter with needle pointing to 10)

10 with probability 100%

x	f(x)
00	0
01	0
10	1
11	0

- **Quantum computing is not about faster hardware. It is about new ways of thinking about computation:** Algorithms exploit superposition, interference, and entanglement to extract global properties of problems.
- **Quantum advantage comes from structure, not brute force:** Algorithms like Simon's, Grover's, and QFT work because the problem has hidden structure (periodicity, symmetry, interference, search space amplification).
- **Measurement is not the end, rather, it is the goal:** Quantum algorithms carefully shape amplitudes so that when we finally measure, the correct answers appear with high probability.
- **The theory is beautiful, but real hardware is imperfect:** Noise, decoherence, and limited qubit counts mean today's machines (NISQ devices) can only run small or approximate algorithms.
- **The field is still evolving, and the biggest opportunities are ahead:** Future progress will require advances not only in algorithms, but also in materials, control systems, error correction, and large-scale engineering.